

$n \rightarrow$ polytropic index

013-830

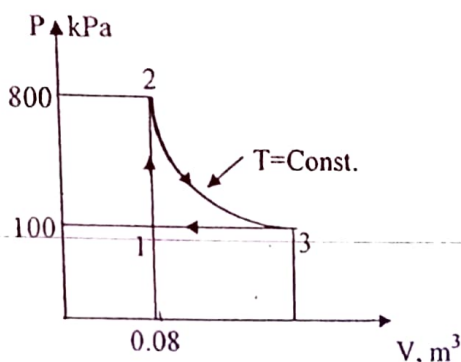
Heat & Work

$$pV^n = \text{const.}$$

8, 14, 21 - 42, 100 - 1200

1. A gas expands from an initial state where $P_1=550$ kPa and $V_1=0.1$ m³ to final state where $P_2=1000$ kPa. The relationship between pressure and volume during the process is $PV^2=\text{Constant}$, determine the work done in kJ. [P.U. 2008Fall/2004Spring]
2. A mass of gas is compressed in a quasi-static process from 80kPa, 0.1 m³ to 0.4 MPa, 0.03m³. Assuming that the pressure and volume are related by $PV^n=\text{constant}$. Find the work done by the gas system. [2007 Spring]
3. For the cycle shown in the figure, determine the work output and the heat transfer.

[PU 2007 Fall]



4. A mass of gas is compressed in a quasi-static process from pressure of 72 kPa and volume of 0.1 m³ to pressure of 0.36 MPa and volume of 0.03 m³. If the pressure and volume are related by $PV^n=\text{constant}$, find the work done by the gas system. [PU 2006 Fall]
5. A gas expands from an initial state where $P_1=500$ kPa and $V_1=0.1$ m³ to a final state where $P_2=100$ kPa. The relationship between pressure and volume during the process is $PV=\text{constant}$. Determine the work in kJ. [PU2006 Spring]
6. A gas is compressed from $V_1=0.09$ m³, $P_1=1$ bar to $V_2=0.03$ m³, $P_2=3$ BAR. Pressure and volume are related linearly during the process. For the gas, find the work, in kJ. [PU 2009 Spring].
7. In a non flow process, a gas expands from volume 0.5 m³ to a volume of 1 m³ according to the law $P=3V^2+1/V$, where P is the pressure in bar and V is the volume in m³. Determine (i) the pressure at the end of the expansion and (ii) the work done by the gas in the expansion process, in kJ. [PU 2010 Fall]
8. 0.2 m³ of an ideal gas at a pressure of 2 MPa and 600K is expanded isothermally to 5 times the initial volume. It is then cooled to 300K at constant volume and then ~~compressed back polytropically to its initial state. Determine the total work done during the cycle.~~
9. A cylinder fitted with a movable piston contains 0.04m³ of air at 10bar pressure and 400K temperature. The air expands according to the law $p = \frac{A}{V^2} - \frac{B}{V}$ to a final pressure of 1 bar and volume of 0.2m³. Determine the work done.
10. A Cylinder fitted with a piston contains propane gas at 100 kPa, 300K with a volume of 0.2m³. The gas is now slowly compressed according to the relation $PV^{1.1}=\text{Const}$ to a final temperature of 340K.
 - a) What is the final pressure?
 - b) How much work is done during the process

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11. 3 kg of air kept at an absolute pressure of 100kPa and temperature of 300K is compressed polytropically until the pressure and temperature becomes 1500kPa and 500K respectively. Evaluate the index of compression and the work of compression. Take $R=287 \text{ J/kg K}$
12. A piston-cylinder arrangement contains carbon dioxide at 300kPa, 100°C with a volume of 0.2 m^3 . Weights are added to the piston such that the gas compresses according to the law $PV^{1.2}=\text{constant}$ to a final temperature of 200°C . Determine the work done during the process.
13. A gas undergoes a thermodynamic cycle consisting of three processes:
Process 1-2: compression with $pV=\text{constant}$, from $p_1=1 \text{ bar}$, $V_1=1.6 \text{ m}^3$ to $V_2=0.2 \text{ m}^3$; $U_2-U_1=0$; Process 2-3: constant pressure to $V_3=V_1$; Process 3-1: constant volume, $U_1-U_3=-3549 \text{ kJ}$. Determine the net work for the cycle.
14. A cylinder fitted with a movable piston contains 9 m^3 of air at 1 bar and 400 K. The air expands according to the law $V=Ap^4+Bp^3$ to a final pressure of 2 bar when the volume is 112 m^3 . Calculate the work done during the expansion.
15. A non-flow reversible process occurs for which pressure and volume are correlated by the expression $p=\left(V^2+\frac{6}{V}\right)$ where p is in bar and V is in m^3 . What amount of work will be done when volume changes from 2 to 4 m^3 ?

HEAT AND WORK

Answers

1. - 19.16 kJ
2. - 11.877 kJ
3. 77.084 kJ
4. - 10.689 kJ
5. 80.47 kJ
6. - 12 kJ
7. 4 bar, 156.8 kJ
8. 179.74 kJ
9. 44.14 kJ
10. a) 0.396 MPa b) - 26.67 kJ
11. 1.23, - 748.7 kJ
12. - 80.4 kJ
13. 787.29 kJ
14. 169 J
15. $22.83 \times 10^3 \text{ J}$

Heat & Work

- (1) A gas expands from an initial state where $P_1 = 550 \text{ kPa}$ and $V_1 = 0.1 \text{ m}^3$ to final state where $P_2 = 1000 \text{ kPa}$. The relationship between pressure and volume during the process is $PV^2 = \text{const}$. determine the work done in kJ. [PV. 2008 Fall / 2009 Sp.]

Sol::-

$$P_1 = 550 \text{ kPa}$$

$$V_1 = 0.1 \text{ m}^3$$

$$P_2 = 1000 \text{ kPa}$$

$$PV^2 = \text{const.}$$

$$n = 2$$

Now,

$$P_1 V_1^2 = P_2 V_2^2$$

$$V_2^2 = \frac{P_1 V_1^2}{P_2}$$

$$V_2 = \sqrt{\frac{P_1 V_1^2}{P_2}}$$

$$= \sqrt{\frac{550 \times (0.1)^2}{1000}}$$

$$\therefore V_2 = 7.416 \times 10^{-2} \text{ m}^3$$

Again,

$$\therefore W_2 = \frac{P_2 V_2 - P_1 V_1}{1-n}$$

$$= \frac{1000 \times 7.416 \times 10^{-2} - 550 \times 0.1}{1-2}$$

$$= -19.16 \text{ kJ} \quad \text{Ans}$$

(2) A mass of gas is compressed in a quasi-static process from 80 kPa, 0.1 m^3 to 0.4 MPa , 0.03 m^3 . Assuming the pressure and volume are related by $pV^n = \text{constant}$. Find the work done by the gas system. [2007 Spring]

Solⁿ:-

$$P_1 = 80 \times 10^3 \text{ Pa}$$

$$V_1 = 0.1 \text{ m}^3$$

$$P_2 = 0.4 \times 10^6 \text{ Pa}$$

$$V_2 = 0.03 \text{ m}^3$$

Now, $P_1 V_1^n = P_2 V_2^n$

$$\left(\frac{V_1}{V_2}\right)^n = \frac{P_2}{P_1}$$

Taking log on both sides

$$n \ln \left(\frac{V_1}{V_2}\right) = \ln \left(\frac{P_2}{P_1}\right)$$

$$n = \frac{\ln (P_2/P_1)}{\ln (V_1/V_2)}$$

$$= \frac{\ln (0.4 \times 10^6 / 80 \times 10^3)}{\ln (0.1 / 0.03)}$$

$$= \frac{\ln (5)}{\ln (10/3)}$$

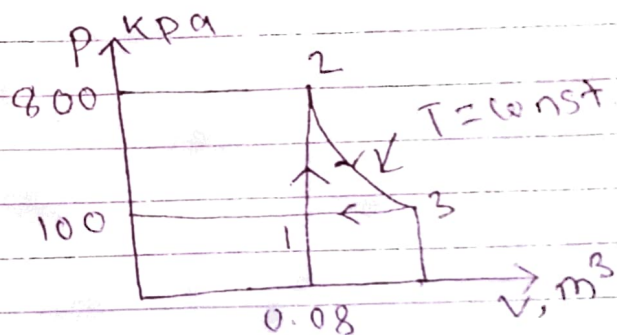
$$= \frac{1.6094}{1.2039}$$

$$= 1.3368$$

$$\therefore n = 1.3368$$

$$\begin{aligned}
 W_2 &= \frac{P_2 V_2 - P_1 V_1}{1 - \gamma} \\
 &= \frac{0.4 \times 10^6 \times 0.03 - 80 \times 10^3 \times 0.1}{1 - 1.3368} \\
 &= \frac{12000 - 8000}{0.3368} \\
 &= -11876.484 \text{ J} \\
 &= -11.876 \text{ kJ} \text{ rev}^{-1}
 \end{aligned}$$

- 3) For the cycle shown in the figure, determine the work output and the heat transfer.



Solⁿ:- From P-V diagram,

$$\begin{aligned}
 P_1 &= 100 \times 10^3 \text{ Pa} \\
 P_3 &= 100 \times 10^3 \text{ Pa} \\
 V_1 &= 0.08 \text{ m}^3 \\
 V_2 &= 0.08 \text{ m}^3 \\
 P_2 &= 800 \times 10^3 \text{ Pa}
 \end{aligned}$$

Now, process 2-3

$T = \text{const.}$

$PV = \text{const.}$

$$P_2 V_2 = P_3 V_3$$

$$V_3 = \frac{P_2 V_2}{P_3}$$

$$= \frac{800 \times 10^3 \times 0.08}{100 \times 10^3}$$

$$\therefore V_3 = 0.64$$

$$2W_3 = P_2 V_2 \ln \left(\frac{V_3}{V_2} \right)$$

$$= 800 \times 10^3 \times 0.08 \ln \left(\frac{0.64}{0.08} \right)$$

$$= 133.084 \text{ kJ}$$

process 1-2.
 $v = \text{const.}$

$$1W_2 = 0$$

process 3-1.

$$3W_1 = P (V_1 - V_3)$$

$$= 100 (0.08 - 0.64)$$

$$= 100 \times (-0.56)$$

$$= -56 \text{ kJ}$$

$$W_{\text{net}} = 0 + 133.084 + (-56)$$

$$= 77.084 \text{ kJ} \quad \underline{\underline{\text{Ans}}}$$

(4) A mass of gas is compressed in a quasi-static process from pressure of 72 kPa and volume of 0.1 m^3 to pressure of 0.36 MPa and volume of 0.03 m^3 . If the pressure and volume are related by $PV^n = \text{constant}$, Find the work done by the gas system. [PU 2006 Fall]

Solⁿ:- $P_1 = 72 \text{ kPa} = 72 \times 10^3 \text{ Pa}$

$$V_1 = 0.1 \text{ m}^3$$

$$P_2 = 0.36 \times 10^6 \text{ Pa}$$

$$V_2 = 0.03 \text{ m}^3$$

Now,

$$P_1 V_1^n = P_2 V_2^n$$

$$\left(\frac{V_1}{V_2} \right)^n = \frac{P_2}{P_1}$$

Taking log on both sides,

$$n \ln \left(\frac{V_1}{V_2} \right) = \ln \left(\frac{P_2}{P_1} \right)$$

$$n = \frac{\ln \left(\frac{P_2}{P_1} \right)}{\ln \left(\frac{V_1}{V_2} \right)}$$

$$= \frac{\ln \left(\frac{0.36 \times 10^6}{72 \times 10^3} \right)}{\ln \left(\frac{0.1}{0.03} \right)}$$

$$= \frac{1.6094}{1.2039}$$

$$= 1.3368$$

$$\therefore W_2 = \frac{P_2 V_2 - P_1 V_1}{1 - \gamma}$$

$$= \frac{0.36 \times 10^6 \times 0.03 - 72 \times 10^3 \times 0.1}{1 - 1.3368}$$

$$= \frac{-3600}{0.3368} \text{ J}$$

$$= -10.689 \text{ kJ} \quad \underline{\text{Ans}}$$

- (5) A gas expands from an initial state where $P_1 = 500 \text{ kPa}$ and $V_1 = 0.1 \text{ m}^3$ to a final state where $P_2 = 100 \text{ kPa}$. The relationship between pressure and volume during the process is $PV = \text{constant}$. Determine the work in kJ. [PU 2006 Spring]

Sol:- $P_1 = 500 \times 10^3 \text{ Pa}$
 $V_1 = 0.1 \text{ m}^3$
 $P_2 = 100 \times 10^3 \text{ Pa}$

Now,

$$PV = \text{constant}$$

$$P_1 V_1 = P_2 V_2$$

$$V_2 = \frac{P_1 V_1}{P_2}$$

$$= \frac{500 \times 10^3 \times 0.1}{100 \times 10^3}$$

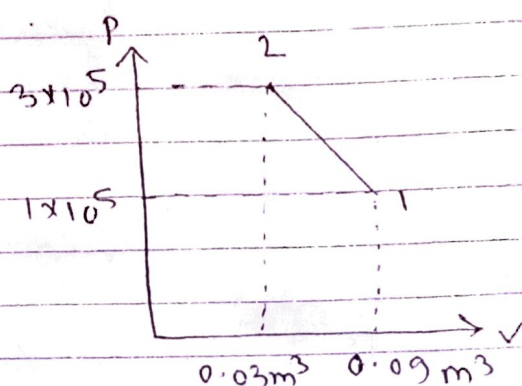
$$\therefore V_2 = 0.5 \text{ m}^3$$

$$\begin{aligned}
 W_2 &= P_1 V_1 \ln \left(\frac{V_2}{V_1} \right) \\
 &= 500 \times 10^3 \times 0.1 \ln \left(\frac{0.5}{0.1} \right) \\
 &= 80.47 \text{ kJ} \text{ Ans.}
 \end{aligned}$$

- (6) A gas is compressed from $V_1 = 0.09 \text{ m}^3$, $P_1 = 1 \text{ bar}$ to $V_2 = 0.03 \text{ m}^3$, $P_2 = 3 \text{ bar}$. pressure and volume are related linearly during the process. For the gas, Find the work, in kJ. [PU 2009 Spring]

Sol:

We know that the area under p-v diagram gives the magnitude of work done



Work done = Area under p-v diagram

$$= (0.09 - 0.03) \times 1 \times 10^5 + \frac{1}{2} \times 0.06 \times (3 - 1) \times 10^5$$

$$= 12000 \text{ J}$$

$$= 12 \text{ kJ}$$

Since,

this one is compression process, so,

$$W_2 = -12 \text{ kJ} \text{ Ans.}$$

7) In a non flow process, a gas expands from volume 0.5 m^3 to a volume of 1 m^3 according to the law $P = 3V^2 + \frac{1}{V}$, where P is the pressure in bar and V is the volume in m^3 . Determine (i) the pressure at the end of the expansion and (ii) the work done by the gas in the expansion process, in kJ. [PU 2010 Fall]

Soln:-

$$V_1 = 0.5 \text{ m}^3$$

$$V_2 = 1 \text{ m}^3$$

(i) The pressure at the end of the expansion = P_2

$$P_2 = 3V_2^2 + \frac{1}{V_2}$$

$V_2 = 1 \text{ m}$

$$P_2 = 3 \times 1^2 + \frac{1}{1}$$

$$= 3 + 1$$

$$= 4 \text{ bar.}$$

(ii)

$$W_2 = \int_{V_1}^{V_2} P dV$$

$$= \int_{V_1}^{V_2} \left(3V^2 + \frac{1}{V} \right) dV \times 10^5$$

$$= \left[3 \int_{V_1}^{V_2} V^2 dV + \int_{V_1}^{V_2} \frac{1}{V} dV \right] \times 10^5$$

$$= \left[3 \left[\frac{v_2^3}{3} \right]_{v_1}^{v_2} + [\ln(v)]_{v_1}^{v_2} \right] \times 10^5$$

$$= \left[3 \left[\frac{v_2^3}{3} - \frac{v_1^3}{3} \right] + [\ln v_2 - \ln v_1] \right] \times 10^5$$

$$= \left[3 \times \frac{1}{3} [v_2^3 - v_1^3] + \ln \left(\frac{v_2}{v_1} \right) \right] \times 10^5$$

$$= \left[1^3 - (0.5)^3 + \ln \left(\frac{1}{0.5} \right) \right] \times 10^5$$

$$= [1 - 0.125 + 0.6931] \times 10^5$$

$$= 1.5681 \times 10^5 \text{ J}$$

$$= 156.81 \text{ kJ} \text{ Ans}$$

- (8) 0.2 m^3 of an ideal gas at a pressure of 2 MPa and 600 K is expanded isothermally to 5 times its initial volume. It is then cooled to 300 K at constant volume and then compressed back polytropically to its initial state. Determine the total work done during the cycle.

Soln:

$$V_1 = 0.2 \text{ m}^3$$

$$P_1 = 2 \times 10^6 \text{ Pa}$$

$$T_1 = 600 \text{ K}$$

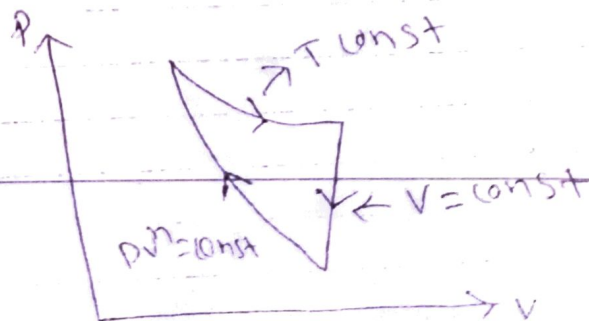
$$V_2 = 5V_1$$

$$= 5 \times 0.2 \text{ m}^3$$

$$= 1 \text{ m}^3$$

$$T_2 = 300 \text{ K} \quad T_1 = 600 \text{ K}$$

$$T_3 = 200 \text{ K}$$



$$V_3 = V_2 = 1 \text{ m}^3$$

Now,

$$\begin{aligned} 1W_2 &= P_1 V_1 \ln \frac{V_2}{V_1} \\ &= 2 \times 10^6 \times 0.2 \times \ln\left(\frac{1}{0.2}\right) \\ &= 643.784 \text{ J} \end{aligned}$$

$$2W_3 = 0 \text{ (Since the process 2-3 is isochoric one)}$$

Now, process 1-2 is isothermal,

$$\begin{aligned} PV &= K \\ P_1 V_1 &= P_2 V_2 \\ P_2 &= \frac{P_1 V_1}{V_2} \\ &= \frac{2 \times 10^6 \times 0.2}{1} \\ &= 0.4 \times 10^6 \text{ Pa} \end{aligned}$$

We know that,

$$PV = nRT$$

when $V = \text{const}$, we have,

$$\frac{P}{T} = \text{const.}$$

Applying this to the process 2-3, we get

$$\frac{P_3}{T_3} = \frac{P_2}{T_2}$$

$$P_3 = P_2 \times \frac{T_3}{T_2}$$

$$= 0.4 \times 10^6 \times \frac{300}{600}$$

$$= 0.2 \times 10^6 \text{ Pa}$$

Now applying $PV^n = K$ to the polytropic process 3-1, we get,

$$P_3 V_3^n = P_1 V_1^n$$

$$\left(\frac{V_3}{V_1} \right)^n = \frac{P_1}{P_3}$$

Taking log on both the sides, we get,

$$\ln \left(\frac{V_3}{V_1} \right)^n = \ln (P_1/P_3)$$

$$\therefore n \ln (V_3/V_1) = \ln (P_1/P_3)$$

$$n = \frac{\ln (P_1/P_3)}{\ln (V_3/V_1)}$$

$$= \frac{\ln (2 \times 10^6 / 0.2 \times 10^6)}{\ln (1/0.2)}$$

$$= 1.431$$

So,

$$\begin{aligned}
 3W_1 &= \frac{P_1 V_1 - P_3 V_3}{1 - n} = \frac{2 \times 10^6 \times 0.2 - 0.2 \times 10^6 \times 1}{1 - 1.431} \\
 &= -464037.123 \text{ J} \\
 &= -464.04 \text{ kJ}
 \end{aligned}$$

∴ Total work done in the cycle is given by,

$$\begin{aligned}
 W_{\text{net}} &= W_2 + 2W_3 + 3W_1 \\
 &= 643.78 + 0 - 464.04 \\
 &= 179.74 \text{ kJ}
 \end{aligned}$$

(9) A cylinder fitted with a movable piston contains 0.04 m^3 of air at 10 bar pressure and 400 K temperature. The air expands according to the law $P = A/\sqrt{V} - B/V$ to a final pressure of 1 bar and volume of 0.2 m^3 . Determine the work done.

Solⁿ:-

$$V_1 = 0.04 \text{ m}^3$$

$$P_1 = 10 \text{ bar}$$

$$T_1 = 400 \text{ K}$$

$$P_2 = 1 \text{ bar}$$

$$V_2 = 0.2 \text{ m}^3$$

By the question, we have

$$P = \frac{A}{\sqrt{V}} - \frac{B}{V}$$

When $P_1 = 10 \text{ bar}$ & $V_1 = 0.04 \text{ m}^3$ we get,

$$10 = \frac{A}{(0.04)^2} - \frac{B}{0.04}$$

$$\{ 10(0.04)^2 = A - B(0.04) \}$$

$$\{ A - 0.04B = 0.16 \text{ --- (i)} \}$$

similarly,

$$\text{When, } P_2 = 1 \text{ bar, } V_2 = 0.2 \text{ m}^3$$

$$1 = \frac{A}{(0.2)^2} - \frac{B}{(0.2)}$$

$$\{ 1(0.2)^2 = A - 0.2B \}$$

$$\{ A - 0.2B = 0.04 \text{ --- (ii)} \}$$

Subtracting eqn (II) from eqn (I), we get

$$A - 0.04B - A + 0.2B = 0.16 - 0.04$$

$$0.16B = -0.024$$

$$\therefore B = \frac{-0.024}{0.16}$$

$$\therefore B = -0.15$$

Putting the value of B in eqn (II), we get,

$$A + 0.2 \times 0.15 = 0.04$$

$$\{ A = 0.04 - 0.03 \}$$

$$\therefore A = 0.01$$

Now,

$$p = \frac{0.01}{v^2} + \frac{0.15}{v} \quad \text{bar}$$

$$= \left[\frac{0.01}{v^2} + \frac{0.15}{v} \right] \times 10^5 \text{ Pa}$$

$$W_2 = \int_{v_1}^{v_2} p \, dv$$

$$= \left[\int_{v_1}^{v_2} \frac{0.01 \, dv}{v^2} + \int_{v_1}^{v_2} \frac{0.15 \, dv}{v} \right] \times 10^5$$

$$= 10^5 \left[0.01 \left| -\frac{1}{v} \right|_{v_1}^{v_2} + 0.15 \left| \ln v \right|_{v_1}^{v_2} \right]$$

$$= 10^5 \left[0.01 \left(-\frac{1}{v_2} + \frac{1}{v_1} \right) + 0.15 (\ln v_2 - \ln v_1) \right]$$

$$= 10^5 \left[0.01 \left(-\frac{1}{0.2} + \frac{1}{0.04} \right) + 0.15 (\ln 0.2 - \ln 0.04) \right]$$

$$= 10^5 \left[0.01 (-5 + 25) + 0.15 (-1.6094 + 3.2189) \right]$$

$$= 10^5 [0.2 + 0.2414]$$

$$= 0.4414 \times 10^5$$

$$= 44.14 \text{ kJ} \quad \underline{\underline{\text{Ans}}}$$

(c) A cylinder fitted with a piston contains propane gas at 100 kPa, 300 K with a volume of 0.2 m^3 . The gas is now slowly compressed according to the relation $pV^{1.1} = \text{const.}$ to a final temperature of 340 K.

(a) What is the final pressure?

(b) How much work is done during the process?

Solⁿ:-

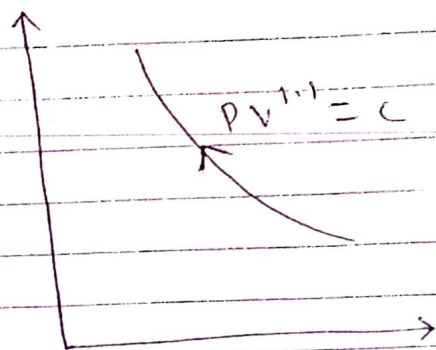
$$P_1 = 100 \text{ kPa}$$

$$T_1 = 300 \text{ K}$$

$$V_1 = 0.2 \text{ m}^3$$

$$T_2 = 340 \text{ K}$$

We know that,



$$\frac{P_2 V_2}{T_2} = \frac{P_1 V_1}{T_1}$$

$$P_2 V_2 = \frac{P_1 V_1 T_2}{T_1}$$

$$= \frac{100 \times 10^3 \times 0.2 \times 340}{300}$$

$$\therefore P_2 V_2 = 22666.67 \text{ --- (i)}$$

Also, by question, we have,

$$pV^{1.1} = c$$

$$P_2 V_2^{1.1} = P_1 V_1^{1.1}$$

$$P_2 V_2^{1.1} = 100 \times 10^3 \times (0.2)^{1.1}$$

$$= 17026.79$$

$$\therefore P_2 V_2^{1.1} = 17026.79 \quad \text{--- (ii)}$$

Dividing (II) by (I) we get,

$$\frac{P_2 V_2^{1.1}}{P_2 V_2} = \frac{17026.79}{22666.67}$$

$$V_2^{0.1} = \frac{17026.79}{22666.67}$$

$$\therefore V_2 = \left(\frac{17026.79}{22666.67} \right)^{1/0.1}$$

$$= 0.0572 \text{ m}^3$$

Now, putting the value of V_2 in eqn (I) we get,

$$P_2 (0.0572) = 22666.67$$

$$\therefore P_2 = \frac{22666.67}{0.0572} = 396270.45 \text{ Pa}$$

$$= 0.396 \text{ MPa} \quad \underline{\text{Ans}}$$

Thus,

work done,

$$W_2 = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

$$= \frac{22666.67 - 100 \times 10^3 \times 0.2}{1 - 1.1}$$

$$= -26666.7 \text{ J}$$

$$= -26.67 \text{ kJ} \quad \text{Ans}$$

11) 3 kg of air kept at an absolute pressure of 100 kPa and temperature of 300 K is compressed polytropically until the pressure and temperature becomes 1500 kPa and 500 K respectively. Evaluate the index of compression and the work of compression. Take $R = 287 \text{ J/kgK}$.

Ans:

$$m = 3 \text{ kg}$$

$$P_1 = 100 \times 10^3 \text{ Pa}$$

$$T_1 = 300 \text{ K}$$

$$P_2 = 1500 \times 10^3 \text{ Pa}$$

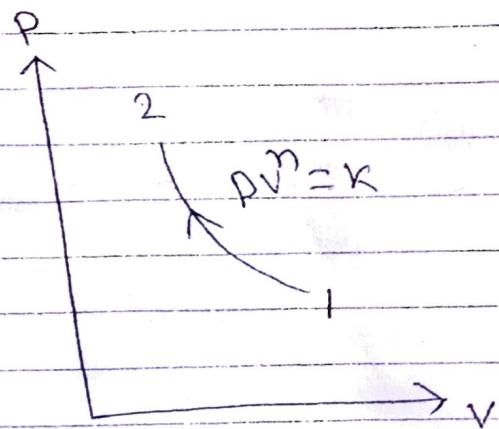
$$T_2 = 500 \text{ K}$$

$$R = 287$$

We know that,

$$P_1 V_1 = m R T_1$$

$$V_1 = \frac{m R T_1}{P_1} = \frac{3 \times 287 \times 300}{100 \times 10^3}$$



$$V_1 = \cancel{0.261} \text{ m}^3 \quad 2.583 \text{ m}^3$$

Also,

$$P_2 V_2 = n R T_2$$

$$V_2 = \frac{n R T_2}{P_2}$$

$$= \frac{3 \times 287 \times 500}{1500 \times 10^3}$$

$$V_2 = 0.287 \text{ m}^3$$

Now,

$$P_2 V_2^n = P_1 V_1^n$$

$$\left(\frac{V_2}{V_1} \right)^n = \frac{P_1}{P_2}$$

Taking log on both sides,

$$n \ln \left(\frac{V_2}{V_1} \right) = \ln \left(\frac{P_1}{P_2} \right)$$

$$n = \frac{\ln \left(\frac{P_1}{P_2} \right)}{\ln \left(\frac{V_2}{V_1} \right)}$$

$$= \frac{\ln \left(\frac{100 \times 10^3}{1500 \times 10^3} \right)}{\ln \left(\frac{0.287}{2.583} \right)}$$

$$= \frac{-27080}{-21972}$$

$$\therefore n = 1.2324 \quad \underline{\text{Ans}}$$

$$\therefore W_2 = \frac{mR(T_2 - T_1)}{1 - n}$$

$$= \frac{3 \times 287 (500 - 300)}{1 - 1.2324}$$

$$= -740.96 \text{ kJ} \quad \underline{\text{Ans}}$$

- (12) A piston-cylinder arrangement contains carbon dioxide at 300 kPa, 100°C with a volume of 0.2 m³. Weights are added to the piston such that the gas compresses according to the law $pV^{1.2} = \text{constant}$ to a final temperature of 200°C. Determine the work done during the process.

Sol:

$$P_1 = 300 \times 10^3 \text{ Pa}$$

$$T_1 = 373 \text{ K}$$

$$V_1 = 0.2 \text{ m}^3$$

$$pV^{1.2} = \text{const.}$$

$$T_2 = 473 \text{ K}$$

We know that,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\begin{aligned}
 P_2 V_2 &= \frac{P_1 V_1 T_2}{T_1} \\
 &= \frac{300 \times 10^3 \times 0.2 \times 473}{373} \\
 &= 76085.790
 \end{aligned}$$

$$\begin{aligned}
 \therefore W_2 &= \frac{P_2 V_2 - P_1 V_1}{1 - \gamma} \\
 &= \frac{76085.790 - 300 \times 10^3 \times 0.2}{1 - 1.2} \\
 &= \frac{76085.790 - 60000}{-0.2} \\
 &= -80.42 \text{ kJ} \quad \text{Ans}
 \end{aligned}$$

- (13) A gas under goes a thermodynamic cycle consisting of three processes:
 process 1-2: compression with $PV = \text{constant}$,
 from $P_1 = 1 \text{ bar}$, $V_1 = 1.6 \text{ m}^3$ to $V_2 = 0.2 \text{ m}^3$;
 $U_2 - U_1 = 0$; process 2-3: - constant pressure
 to $V_3 = V_1$; process 3-1: constant volume
 $U_1 - U_3 = -3549 \text{ kJ}$. Determine the net
 work for the cycle.

Soln:-

$$P_1 = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$$

$$V_1 = 1.6 \text{ m}^3$$

$$V_2 = 0.2$$

$$V_3 = 1.6 \text{ m}^3$$

$${}_1W_2 = P_1 V_1 \ln(V_2/V_1)$$

$$= 1 \times 10^5 \times 1.6 \ln(0.2/1.6)$$

$$= -332710.64$$

For process, 1-2,

$$P_1 V_1 = P_2 V_2$$

$$P_2 = \frac{P_1 V_1}{V_2}$$

$$= \frac{1 \times 1.6}{0.2}$$

$$= 8 \text{ bar}$$

$${}_2W_3 = P_2 (V_3 - V_2)$$

$$= 8 \times 10^5 (1.6 - 0.2)$$

$$= 1120000$$

$${}_3W_1 = 0 \text{ [volume is constant]}$$

$$W_{\text{net}} = -332710.64 + 1120000$$

$$= 787289.36 \text{ J}$$

(14) A cylinder fitted with a movable piston contains 9 m^3 of air at 1 bar and 400 K . The air expands according to the law $V = Ap^4 + Bp^3$ to a final pressure of 2 bar when the volume is ~~at~~ 112 m^3 . Calculate the work done during the expansion.

Sol:

$$V_1 = 9 \text{ m}^3$$

$$P_1 = 1 \text{ bar}$$

$$T_1 = 400 \text{ K}$$

$$P_2 = 2 \text{ bar}$$

$$V_2 = 112 \text{ m}^3$$

When, $P_1 = 1 \text{ bar}$ & $V_1 = 9$

$$9 = A + B$$

$$A + B = 9 \quad \text{--- (i)}$$

When,

$$P_2 = 2 \text{ bar} \text{ \& } V_2 = 112 \text{ m}^3$$

$$112 = 16A + 8B \quad \text{--- (ii)}$$

Solving (i) & (ii) we get,

$$A = 5$$

$$B = 4$$

$$V = Ap^4 + Bp^3$$

$$dV = (4Ap^3 + 3Bp^2) dp$$

$$= (20p^3 + 12p^2) dp$$

$$W_2 = \int_1^2 P dV$$

$$= \int_1^2 P(20P^3 + 12P^2) dP \times 10^5$$

$$= \left[\int_1^2 20P^4 dP + \int_1^2 12P^3 dP \right] \times 10^5$$

$$= \left[20 \frac{P^5}{5} \right]_1^2 + \left[12 \frac{P^4}{4} \right]_1^2$$

$$= 4 [P^5]_1^2 + 3 [P^4]_1^2$$

$$= 4 [32 - 1] + 3 [16 - 1] \times 10^5$$

$$= [4 \times 31 + 3 \times 15] \times 10^5$$

$$= 169 \text{ J} \text{ ~~Ans~~$$

$$= 169 \times 10^5 \text{ J} \text{ Ans$$

(15) A non-flow reversible process occurs for which pressure and volume are connected by the expression $p = \left(v^2 + \frac{6}{v} \right)$ where

p is in bar and v is in m^3 . What amount of work will be done when volume changes from 2 to 4 m^3 .

Sol.

$$v_1 = 2 \text{ m}^3$$

$$v_2 = 4 \text{ m}^3$$

$$p = \left(v^2 + \frac{6}{v} \right)$$

$$W_2 = \int p \, dv$$

$$= \int_{v_1}^{v_2} \left(v^2 + \frac{6}{v} \right) dv$$

$$= \int_2^4 v^2 \, dv + 6 \int_2^4 \frac{1}{v} \, dv$$

$$= \left[\frac{v^3}{3} \right]_2^4 + 6 \left[\ln v \right]_2^4$$

$$= \frac{64}{3} - \frac{16}{3} + 6 [\ln 4 - \ln 2]$$

$$= 16 + 6 \times 0.693$$

$$= 20.158 \text{ J m}^3$$

Prabin choudhary

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